

Coringtreepotters ms outline

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Introduction

1. Climate change shift plant phenology
 - (a) Earlier spring and delayed autumn lengthens the growing season
 - (b) With longer seasons, trees should grow more
 - (c) Temperate forests offset a large portion of human CO₂ emissions
 - (d) Most of that carbon is stored in mature, canopy trees so they are important
2. Increased growth with longer seasons may not happen
 - (a) Some new evidence shows trees don't grow more with longer seasons
 - (b) This could mislead carbon models
 - (c) Highlights the need to understand why trees might not grow more
3. Growth strategies (i.e. acquisitive vs conservative) and wood anatomy may guide species responses to GS
 - (a) Diffuse vs ring porous species start growth at different time—experience different temperature throughout the season
 - (b) Longer seasons may benefit slow growth species > fast growth species: slow growth rate under good conditions for a long time = increased growth
 - (c) Different carbon allocation strategies may mute short term growth responses: storage investment vs opportunistic growth
 - (d) Increased C storage in current year, but stored instead of invested in growth: importance of considering multiple years
4. Previous year conditions can affect the following year's growth
 - (a) Deciduous trees use stored carbon at the beginning of the season
 - (b) Mature trees especially are known to rely on stored carbon for early growth
 - (c) Thus, effect of environmental conditions on yearly growth can be blurred by a multi-year lag effect
5. Look at growth and GSL relationship in an urban arboretum where drought and competition are limited
 - (a) Will look at 11 species of mature trees across nine years.
 - (b) GS number of days vs temperature conditions
 - (c) Growth with tree rings and SOS and EOS with leaf phenology from citizen scientists

Methods

We did a bunch of stuff that I am not outlining. Phenophases by (Denny *et al.*, 2014)

Results

1. Summary basic information

- (a) Dataset includes 11 deciduous tree species with leaf phenology data spanning nine years (2016 to 2024) for a total of 321 leafout and 284 leaf coloring observations, yielding 278 complete growing seasons (GS) for 49 individuals.
- (b) Mean summer temperatures spanned 21.53°C (2017) to 23.07°C (2020), with an average of 22.51°C across the nine years.
- (c) 2020 and 2022 had moderate droughts in the summer (June, July and August) ramped up to severe in September. A moderate drought in June 2016, ramped up to severe in July, then extreme in August and September.
- (d) GSL ranged from 77 days (*A. flava* in 2018) to 194 days (*A. saccharum* in 2021)
- (e) GS thermal sum ranged from 1048.8 GDD (*A. flava* in 2020) to 2634.7 GDD (*T. americana* in 2016)
- (f) The earliest leafout day of year was 08-Apr (*A. flava* in 2021) and the latest, 15-May (*C. ovata* in 2020). The earliest leaf coloring day of year was 15-Jul (*A. flava* in 2021—same individual with earliest leafout) and the latest, 06-Nov (*T. americana* in 2016)
- (g) Annual ring widths across years varied from 0.43 mm (*A. saccharum* in 2019) to 11.75 mm (*T. americana* in 2019) (Figure S2).

2. Does tree growth increase with longer growing seasons?

- (a) No clear trend of growing season length effect across our eleven species and season predictors.
- (b) Results reported as means and 90% uncertainty intervals (UI) that correspond to ring width change in mm on the log scale—henceforth "RW" per scaled predictor (one week (seven days) of average spring growing degree days (GDD); (58.32) for the thermal season, one week of growing season length (GSL) for the calendar season, one week of leafout for the start of season (SOS) and one week of of budset for the end of season (EOS).
- (c) Longer thermal season did not increase growth for any species and decreased growth for *B. alleghaniensis* (-0.08, UI: -0.12, -0.03 RW change per scaled GDD; Figure 1a and e; Table S1).
- (d) No species responded to longer calendar season (Figure 1b and f; Table S1).
- (e) On a scaled axis (z-scored), both GDD and GSL were similarly predictive (Figure S10)

3. Does a warmer thermal season increase growth in the following year?

- (a) Previous year GDD increased growth for *B. alleghaniensis* (0.15, UI: 0.02, 0.28 RW change per scaled GDD), and decreased growth for *B. nigra* (-0.08, UI: -0.17, 0 RW change per scaled GDD; Figure 2b)
- (b) Interestingly, *B. nigra* did not solely respond to current year's conditions (Figure 1a and e; Table S1), but grew more with warmer thermal season when the previous conditions were considered (0.12, UI: 0.03, 0.21 RW change per scaled GDD; Figure 2a).

4. Did an earlier start and later end of season increase growth?

- (a) An earlier start of season did not increase growth for any species and decreased growth for *T. americana* (0.15, UI: 0.02, 0.27 RW change per seven SOS days; Figure 1c and g; Table S1), which did not respond to the thermal or the calendar season.
- (b) A later end of season did not increase growth for any species and decreased growth for *B. alleghaniensis* (-0.09, UI: -0.16, -0.01 RW change per seven EOS days; Figure 1d and h; Table S1), which also decreased growth with warmer thermal season.
- (c) The estimates were unchanged when using the full or restricted datasets (SOS $n = 321$ and EOS $n = 284$ vs $n = 278$; Figure S8b), or when using budburst instead of leafout as the start of the growing season (Figure S9c).
- (d) A later SOS delayed EOS only for *P. deltoides* (1.04, UI: 0.39, 1.7 EOS change in days per SOS day; Figure S11)

5. How did tree growth change across years and tree ids?

- (a) We did not find any clear effect of year on growth (Figure S1) and no clear differences of growth across species (Figure S5 and Figure S6)
- (b) Growth was similar across individual trees (Figure S7).

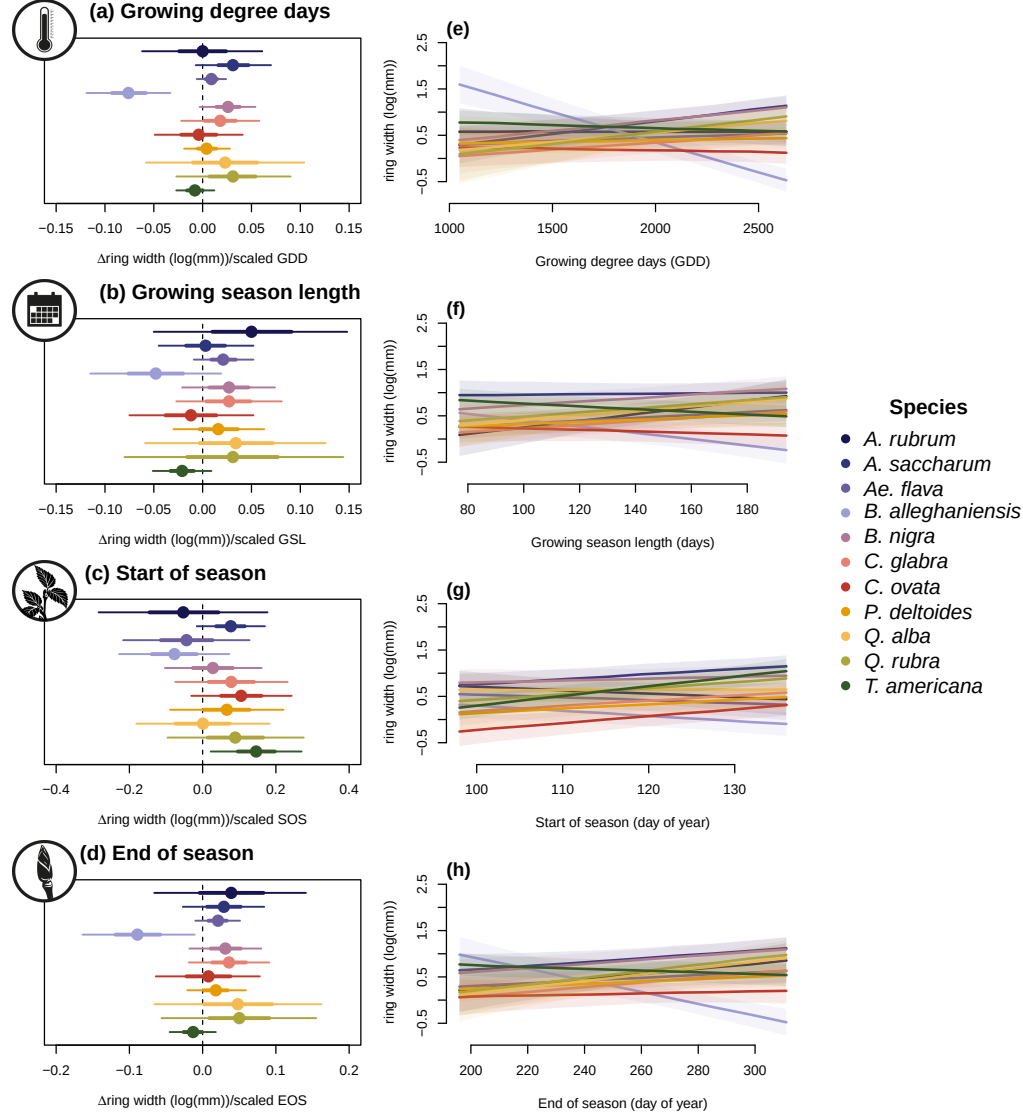
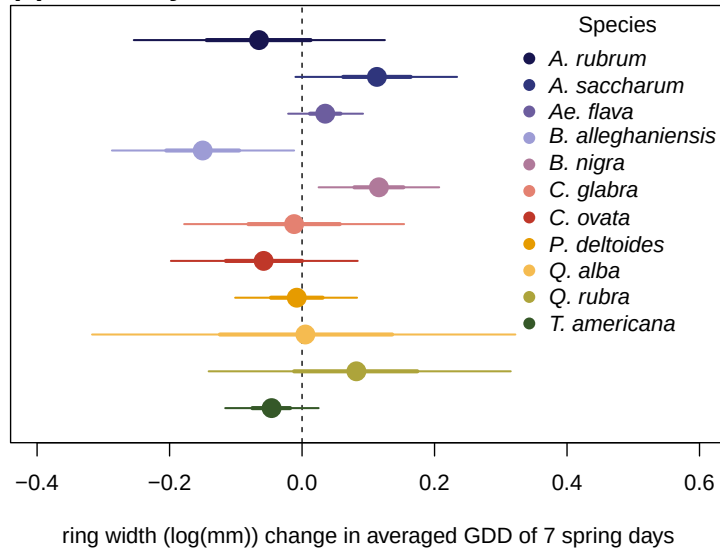


Figure 1: Estimates of ring width change as a function of four different predictors (growing degree days, GDD (mean daily temperature above 5°C); growing season length, GSL; start of season, SOS; end of season, EOS) across our eleven species. The species colors are the same throughout the figures. Panels (a) to (d) represent the model slope estimates for each species, for each predictor, which we represent through ring width change on the log scale per seven average spring days GDD (58.32 GDD; panel a), seven days of GSL (panel b), seven days of leafout (SOS; panel c) and seven days of leaf colouring (EOS; panel d). The dots represent the mean of each species and the lines, the 50% and 90% uncertainty intervals (UI; see Table S1 for parameter estimates). Panels (e) to (h) represent the simulated response curves of ring width as a function of each predictor, with the lines depicting the mean response at each simulated predictor value, and the shaded area, the interquartile ranges (25% and 75% UI). These model do not include a slope on the previous year GDD (see Figure 2 for these estimates).

(a) Current year



(b) Previous year

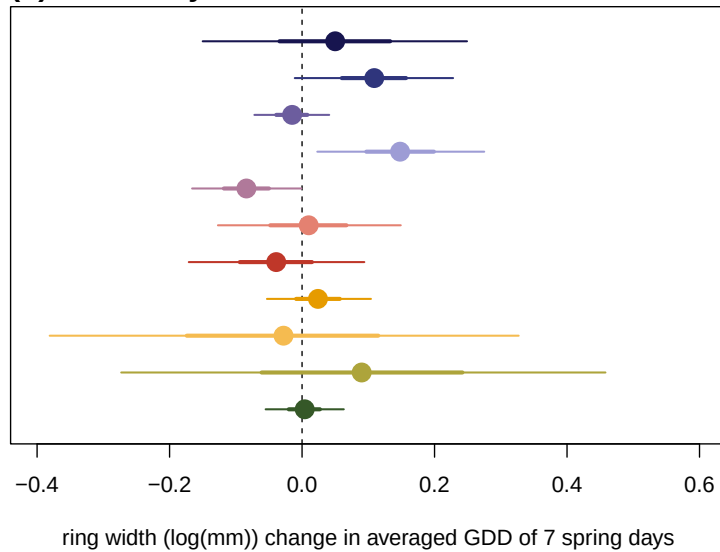


Figure 2: Model slope estimates representing ring width change as a function of the current year growing degree days (GDD; a) and the previous year GDD (b). GDD represent the accumulated growing degree days between leafout and leaf colouring and the models estimates shows ring width change (on the log scale) per seven average spring days GDD (58.32; see data analyses for more details). The dots represent the mean of each species and the lines, the 50% and 90% uncertainty intervals (UI).

Discussion

1. Used 49 trees of 11 species to understand if mature trees grew more with longer GS
 - (a) Annual growth with tree rings and leaf phenology for SOS and EOS
 - (b) Warmer and longer GS did not increase growth for any species and decreased growth for one
 - (c) Previous year conditions shifted growth trend for two species
 - (d) Most species did not respond to SOS or EOS
2. Despite varying growth strategies, no consistent growth trend
 - (a) Thermal or calendar season had no effect on any species except for one
 - (b) *B. alleghaniensis* decrease in growth reveal that this species growth may be inhibited by warmer seasons
 - (c) *B. alleghaniensis* response to thermal season is consequential with more growth with earlier EOS: period of high GDD accumulation
 - (d) *T. americana*: grows less with earlier SOS; shows cool early spring days may reduce growth
 - (e) These suggest no difference of current GS condition/length between acquisitive vs conservative species
3. Previous year conditions affected growth for two *Betula*
 - (a) Diametrically opposed responses between *B. alleghaniensis* and *B. nigra*
 - (b) *B. alleghaniensis* trend suggests that warmer conditions in previous year increases growth, but current decreases
 - (c) Points towards allocation strategy that prioritize storage before growth—expected with conservative species
 - (d) *B. nigra* increases in the current year, but only when previous year is included
 - (e) This highlights the importance of considering not only the current year conditions
4. Weak responses across species corroborates previous findings on mature deciduous trees
 - (a) Our mature trees don't grow more with warmer or longer seasons
 - (b) Considering previous year condition is important, which may blur trees responses to current GS length/conditions
 - (c) GS are expected to keep increasing, but tree growth may not follow
 - (d) Temperate forest carbon sink may not keep increasing as expected

Supplemental results

Table S1: Species level slope parameters which represent ring width change (on the log scale), per scaled predictor (see data analyses section for more details). We report estimates as mean and 90% uncertainty intervals (in square brackets) across our four different predictors (GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season).

Species	GDD	GSL	SOS	EOS
A. rubrum	0.00 [-0.06, 0.06]	0.05 [-0.05, 0.15]	-0.05 [-0.28, 0.18]	0.04 [-0.07, 0.14]
A. saccharum	0.03 [-0.01, 0.07]	0.00 [-0.04, 0.05]	0.08 [-0.02, 0.17]	0.03 [-0.03, 0.08]
Ae. flava	0.01 [-0.01, 0.02]	0.02 [-0.01, 0.05]	-0.04 [-0.22, 0.13]	0.02 [-0.01, 0.05]
B. alleghaniensis	-0.08 [-0.12, -0.03]	-0.05 [-0.12, 0.02]	-0.08 [-0.23, 0.07]	-0.09 [-0.16, -0.01]
B. nigra	0.03 [-0.00, 0.05]	0.03 [-0.02, 0.07]	0.03 [-0.10, 0.16]	0.03 [-0.02, 0.08]
C. glabra	0.02 [-0.02, 0.06]	0.03 [-0.03, 0.08]	0.08 [-0.07, 0.23]	0.04 [-0.02, 0.09]
C. ovata	-0.00 [-0.05, 0.04]	-0.01 [-0.07, 0.05]	0.10 [-0.03, 0.24]	0.01 [-0.06, 0.08]
P. deltoides	0.00 [-0.02, 0.03]	0.02 [-0.03, 0.06]	0.07 [-0.09, 0.22]	0.02 [-0.02, 0.06]
Q. alba	0.02 [-0.06, 0.10]	0.03 [-0.06, 0.13]	0.00 [-0.18, 0.18]	0.05 [-0.07, 0.16]
Q. rubra	0.03 [-0.03, 0.09]	0.03 [-0.08, 0.14]	0.09 [-0.10, 0.28]	0.05 [-0.06, 0.15]
T. americana	-0.01 [-0.03, 0.01]	-0.02 [-0.05, 0.01]	0.15 [0.02, 0.27]	-0.01 [-0.04, 0.02]

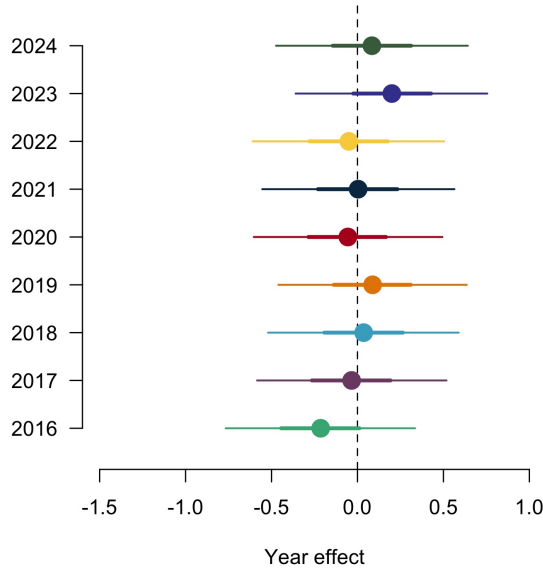


Figure S1: Ring width (on the log scale) deviations from the grand mean for each year using the GDD model (Table S2). The dots are the mean of the posterior distribution and the thick and thin lines, the 50 and 90% UI.

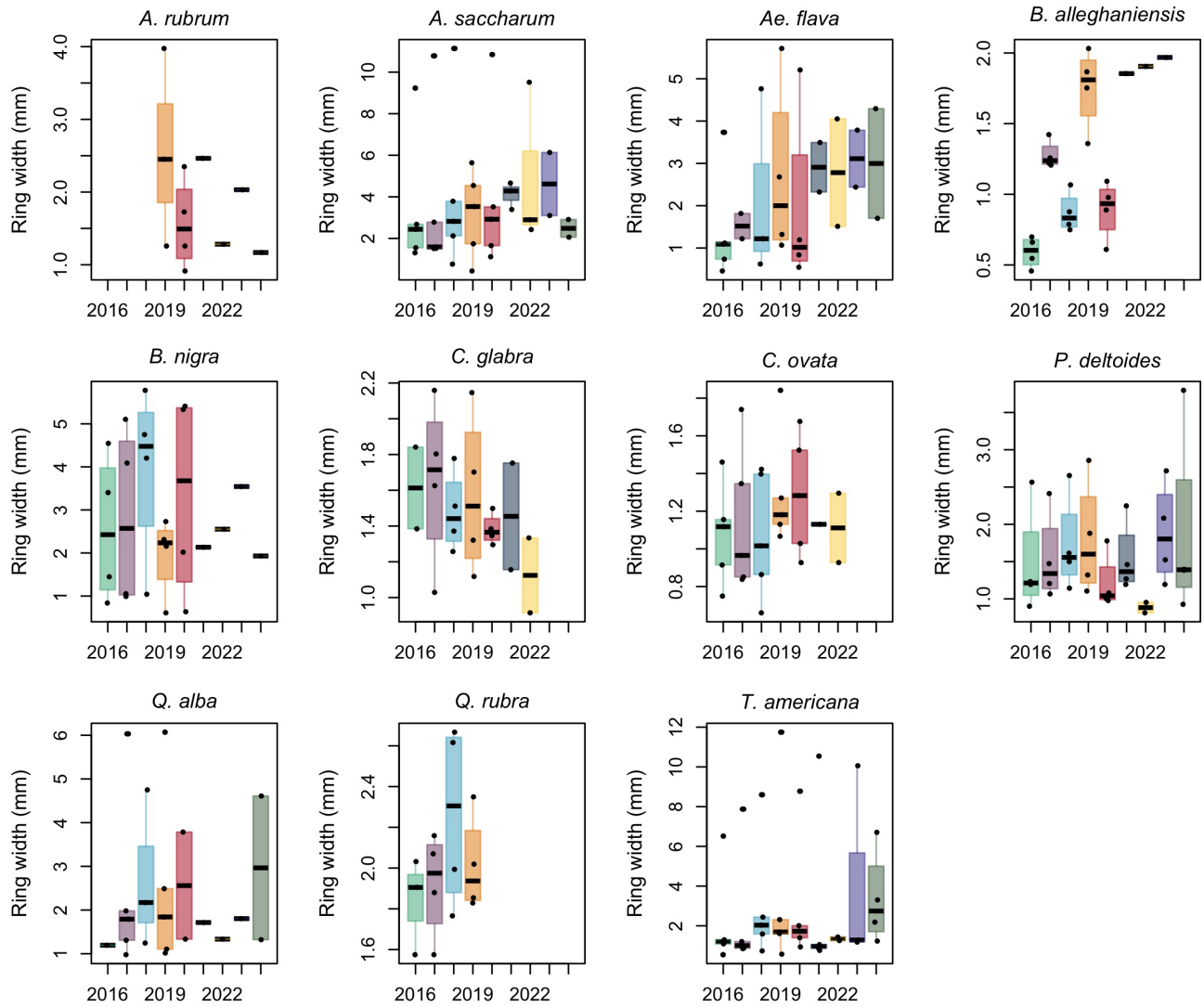


Figure S2: Boxplot representing the empirical data of our ring widths (mm) across years, faceted by species. The dots represent the raw data, the line the mean and the box the interquartile range (IQR).

Table S2: Summary output of each parameter from our four predictors, using ring width (on the log scale) as the reponse variable. We report estimates as mean and 90% uncertainty intervals (in square brackets) across our four different predictors (GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season). See data analyses section for more details.

Parameter	GDD	GSL	SOS	EOS
α	0.61 [-2.08, 3.25]	0.49 [-2.09, 3.03]	0.24 [-2.39, 2.94]	0.23 [-2.48, 2.98]
$\sigma_{\alpha_{tree}}$	0.55 [0.45, 0.67]	0.55 [0.45, 0.68]	0.54 [0.44, 0.66]	0.55 [0.45, 0.67]
σ_y	0.31 [0.29, 0.34]	0.32 [0.29, 0.34]	0.32 [0.29, 0.34]	0.32 [0.29, 0.34]
$\beta_{sppA.rubrum}$	0.00 [-0.06, 0.06]	0.05 [-0.05, 0.15]	-0.05 [-0.28, 0.18]	0.04 [-0.07, 0.14]
$\beta_{sppA.saccharum}$	0.03 [-0.01, 0.07]	0.00 [-0.04, 0.05]	0.08 [-0.02, 0.17]	0.03 [-0.03, 0.08]
$\beta_{sppAe.flava}$	0.01 [-0.01, 0.02]	0.02 [-0.01, 0.05]	-0.04 [-0.22, 0.13]	0.02 [-0.01, 0.05]
$\beta_{sppB.alleganiensis}$	-0.08 [-0.12, -0.03]	-0.05 [-0.12, 0.02]	-0.08 [-0.23, 0.07]	-0.09 [-0.16, -0.01]
$\beta_{sppB.nigra}$	0.03 [-0.00, 0.05]	0.03 [-0.02, 0.07]	0.03 [-0.10, 0.16]	0.03 [-0.02, 0.08]
$\beta_{sppC.glabra}$	0.02 [-0.02, 0.06]	0.03 [-0.03, 0.08]	0.08 [-0.07, 0.23]	0.04 [-0.02, 0.09]
$\beta_{sppC.ovata}$	-0.00 [-0.05, 0.04]	-0.01 [-0.07, 0.05]	0.10 [-0.03, 0.24]	0.01 [-0.06, 0.08]
$\beta_{sppP.deltoides}$	0.00 [-0.02, 0.03]	0.02 [-0.03, 0.06]	0.07 [-0.09, 0.22]	0.02 [-0.02, 0.06]
$\beta_{sppQ.alba}$	0.02 [-0.06, 0.10]	0.03 [-0.06, 0.13]	0.00 [-0.18, 0.18]	0.05 [-0.07, 0.16]
$\beta_{sppQ.rubra}$	0.03 [-0.03, 0.09]	0.03 [-0.08, 0.14]	0.09 [-0.10, 0.28]	0.05 [-0.06, 0.15]
$\beta_{sppT.americana}$	-0.01 [-0.03, 0.01]	-0.02 [-0.05, 0.01]	0.15 [0.02, 0.27]	-0.01 [-0.04, 0.02]
$\alpha_{year2019}$	0.09 [-0.46, 0.64]	0.06 [-0.48, 0.63]	0.07 [-0.47, 0.64]	0.06 [-0.50, 0.62]
$\alpha_{year2020}$	-0.06 [-0.60, 0.49]	-0.07 [-0.61, 0.50]	-0.12 [-0.66, 0.45]	-0.07 [-0.63, 0.47]
$\alpha_{year2021}$	0.00 [-0.55, 0.56]	-0.00 [-0.55, 0.57]	0.01 [-0.54, 0.58]	-0.01 [-0.58, 0.54]
$\alpha_{year2022}$	-0.05 [-0.61, 0.50]	-0.07 [-0.63, 0.50]	-0.07 [-0.63, 0.49]	-0.08 [-0.65, 0.48]
$\alpha_{year2023}$	0.20 [-0.36, 0.76]	0.18 [-0.38, 0.75]	0.20 [-0.35, 0.78]	0.17 [-0.40, 0.72]
$\alpha_{year2024}$	0.08 [-0.47, 0.64]	0.07 [-0.47, 0.66]	0.07 [-0.48, 0.64]	0.08 [-0.49, 0.64]
$\alpha_{year2016}$	-0.21 [-0.77, 0.34]	-0.22 [-0.76, 0.35]	-0.25 [-0.80, 0.31]	-0.25 [-0.81, 0.29]
$\alpha_{year2017}$	-0.03 [-0.58, 0.52]	-0.06 [-0.59, 0.52]	0.00 [-0.54, 0.56]	-0.07 [-0.63, 0.48]
$\alpha_{year2018}$	0.04 [-0.52, 0.59]	0.02 [-0.52, 0.59]	-0.00 [-0.55, 0.56]	0.01 [-0.54, 0.56]
$\alpha_{sppA.rubrum}$	-0.05 [-3.40, 3.35]	-0.95 [-4.12, 2.17]	1.23 [-3.23, 5.60]	-1.08 [-5.62, 3.44]
$\alpha_{sppA.saccharum}$	-0.90 [-3.86, 2.10]	0.43 [-2.34, 3.17]	-0.57 [-3.59, 2.35]	-0.37 [-3.79, 3.02]
$\alpha_{sppAe.flava}$	-0.49 [-3.20, 2.19]	-0.46 [-3.09, 2.14]	0.93 [-2.69, 4.47]	-0.49 [-3.33, 2.38]
$\alpha_{sppB.alleganiensis}$	2.32 [-0.68, 5.34]	0.61 [-2.29, 3.46]	1.17 [-2.34, 4.58]	3.25 [-0.47, 6.96]
$\alpha_{sppB.nigra}$	-0.68 [-3.54, 2.12]	-0.13 [-2.82, 2.58]	0.17 [-3.18, 3.40]	-0.50 [-3.65, 2.67]
$\alpha_{sppC.glabra}$	-0.89 [-3.89, 2.06]	-0.64 [-3.44, 2.12]	-1.17 [-4.65, 2.33]	-1.16 [-4.46, 2.12]
$\alpha_{sppC.ovata}$	-0.32 [-3.35, 2.74]	-0.07 [-2.88, 2.73]	-1.96 [-5.42, 1.40]	-0.35 [-4.01, 3.29]
$\alpha_{sppP.deltoides}$	-0.38 [-3.14, 2.40]	-0.39 [-3.09, 2.26]	-1.04 [-4.60, 2.52]	-0.49 [-3.54, 2.50]
$\alpha_{sppQ.alba}$	-0.85 [-4.70, 3.03]	-0.56 [-3.66, 2.52]	0.40 [-3.47, 4.32]	-1.44 [-6.32, 3.48]
$\alpha_{sppQ.rubra}$	-1.09 [-4.47, 2.23]	-0.43 [-3.80, 2.96]	-1.08 [-4.98, 2.91]	-1.44 [-6.04, 3.22]
$\alpha_{sppT.americana}$	0.30 [-2.41, 3.02]	0.59 [-1.98, 3.18]	-2.02 [-5.24, 1.14]	0.92 [-1.93, 3.81]

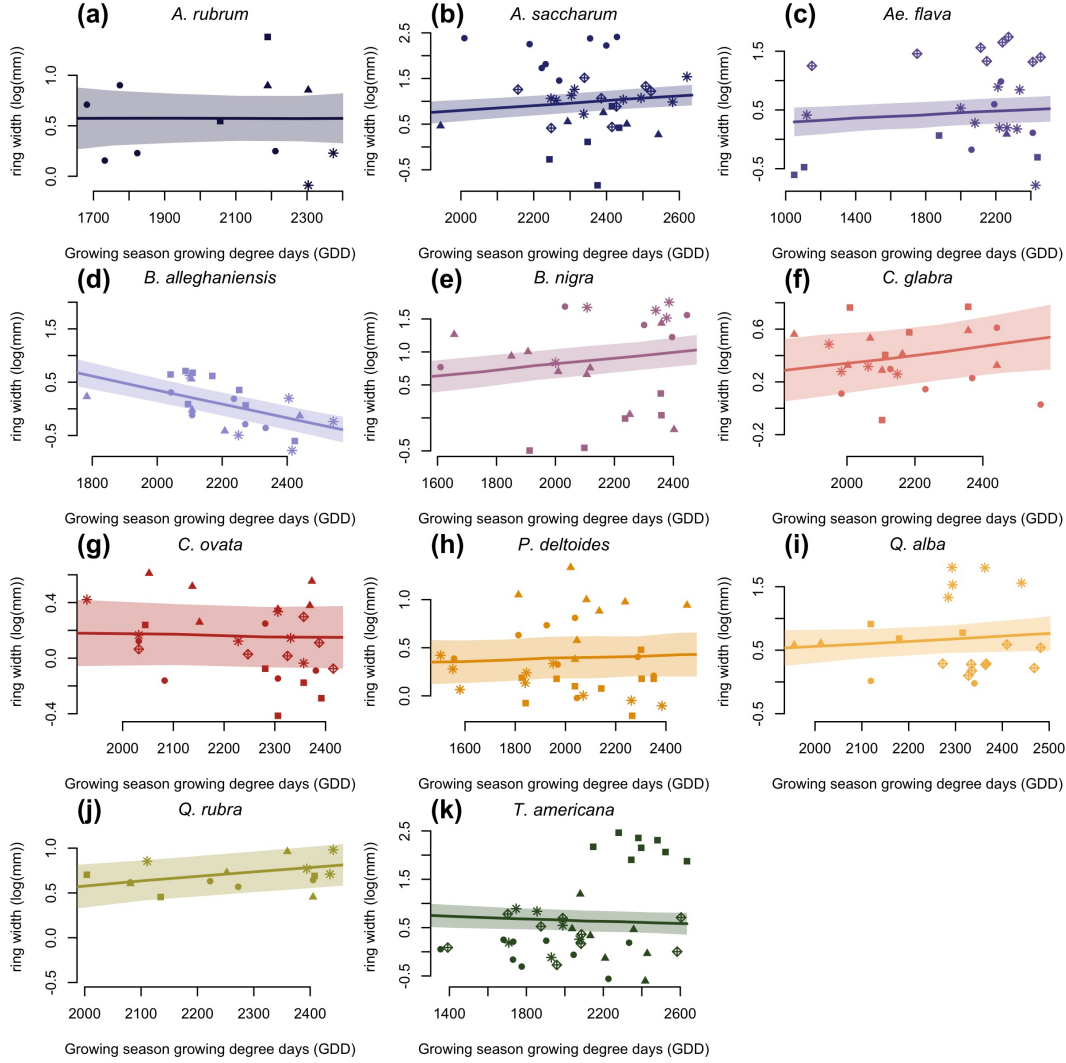


Figure S3: Ring width (on the log scale) trends in response to thermal season (GDD). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by treeids.

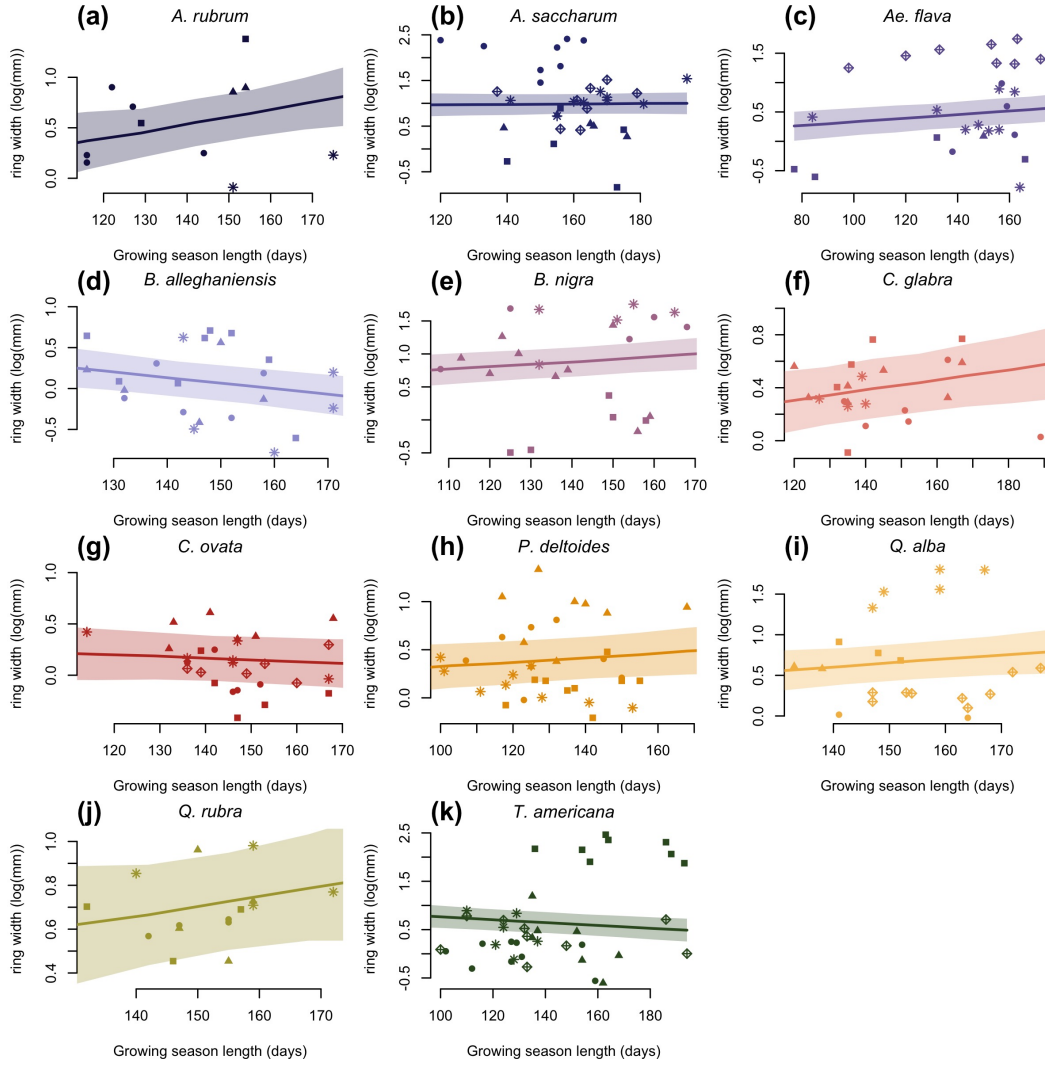


Figure S4: Ring width trends (on the log scale) in response to calendar season (GSL). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by treeids.

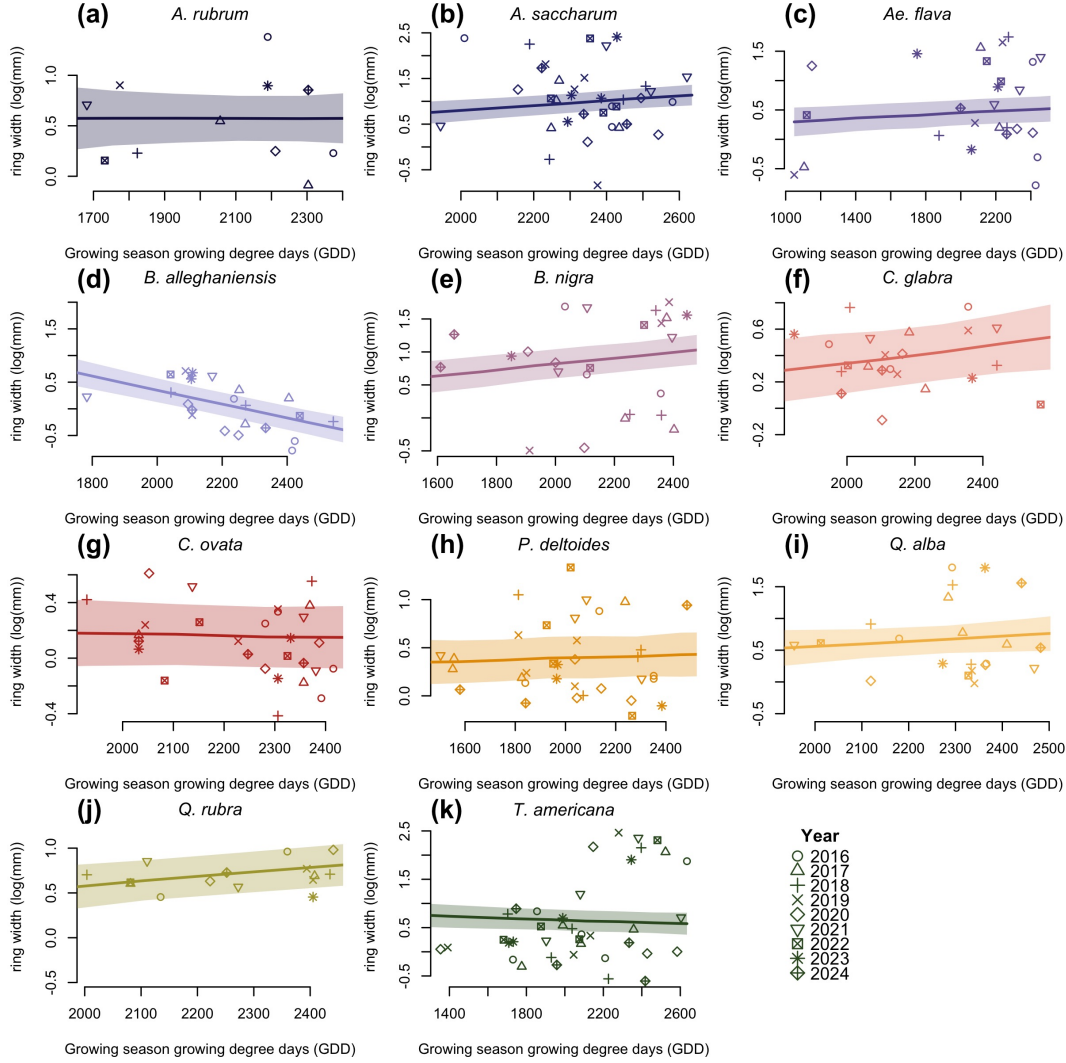


Figure S5: Ring width trends (on the log scale) in response to thermal season (GDD). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by year.

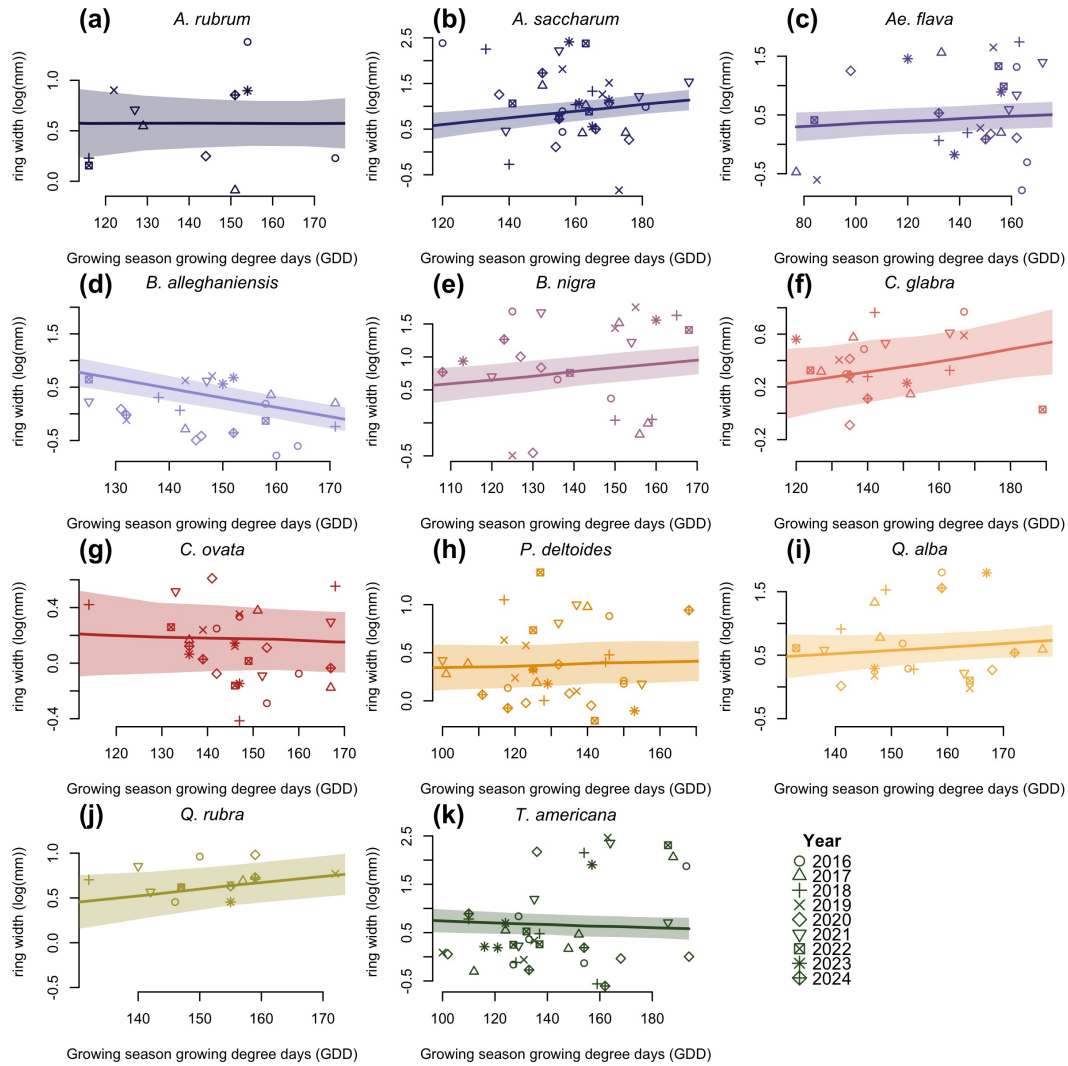


Figure S6: Ring width trends (on the log scale) in response to calendar season (GSL). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by year.

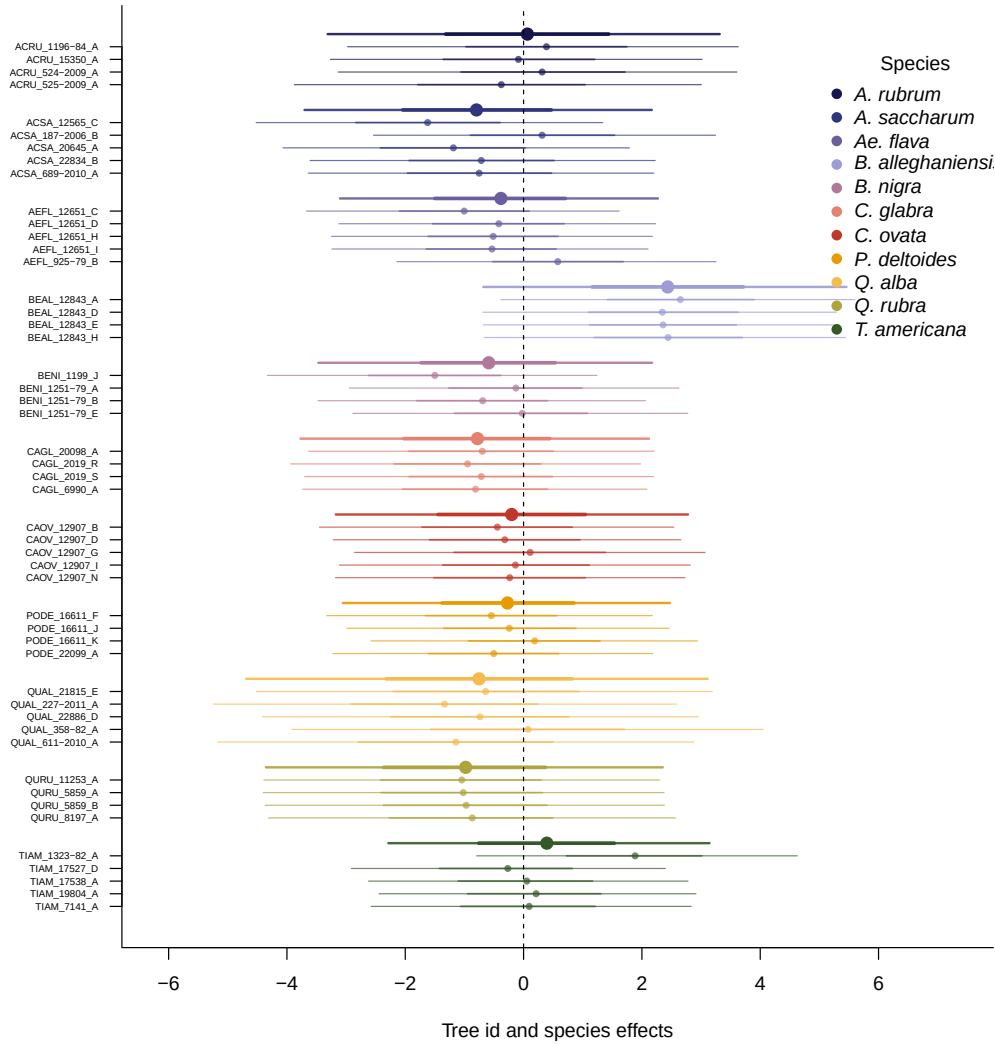
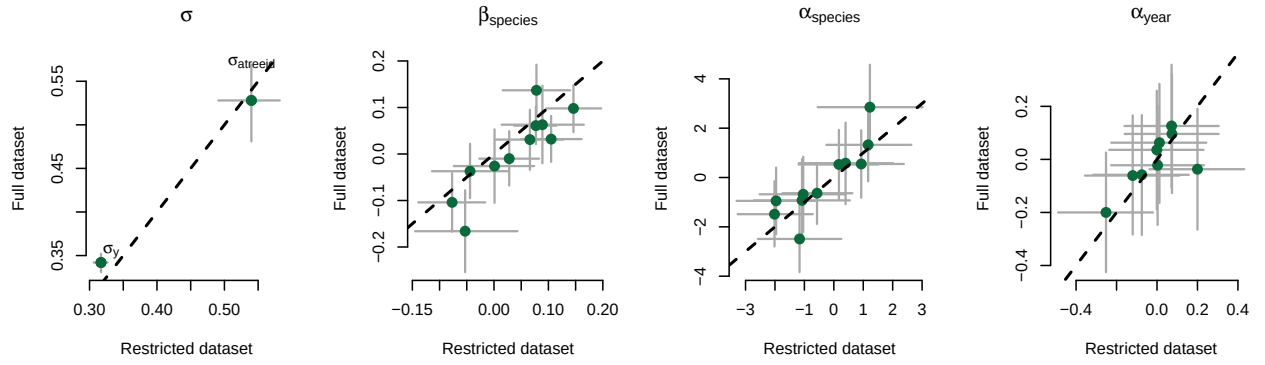


Figure S7: Ring width (on the log scale) deviations from the grand mean for each grouping factor of the GDD model. The different tree id values represent the combined intercepts of tree id, species and averaged year intercepts. The larger dots and lines represent the species intercepts which are the species level estimates averaged across all of their tree ids. The dots are the mean of the posterior distribution and the thick and thin lines, the 50 and 90% UI.

(a) Start of season (SOS)



(b) End of season (EOS)

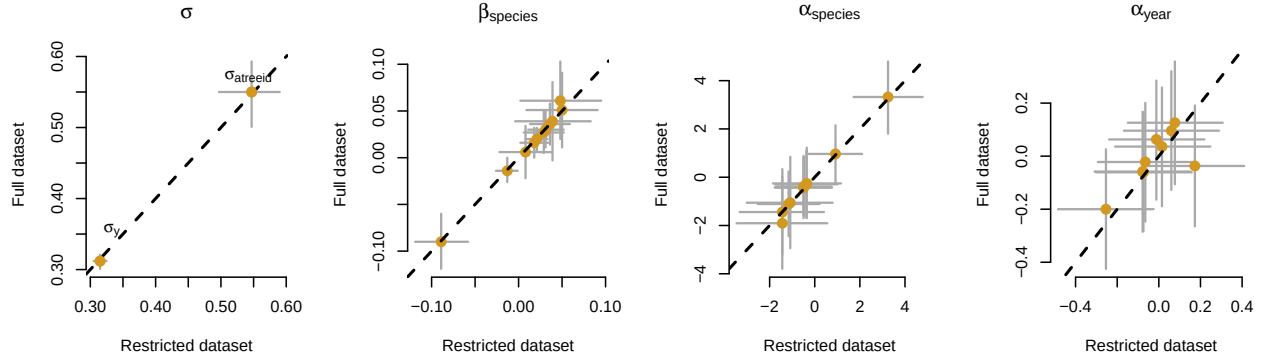


Figure S8: Full vs most restricted rows of data for the model with SOS (a) EOS (b) as the predictor. Each pannel represent the different parameters used to fit the models (see data analyses for more details).

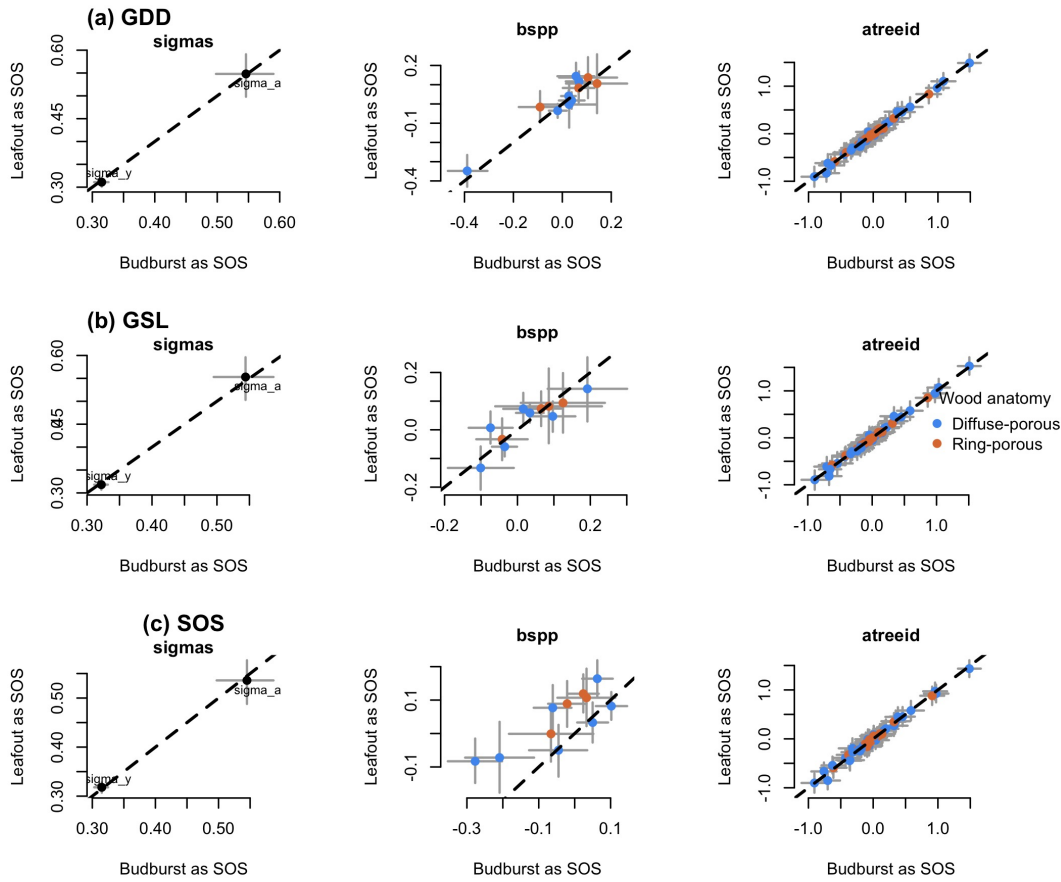


Figure S9: Parameter estimates when calculating the GDD (a), GSL (b) from leafout vs budburst, to leaf coloring. (c) shows the model estimates when taking leafout vs budburst as the SOS (see data analyses for more details).

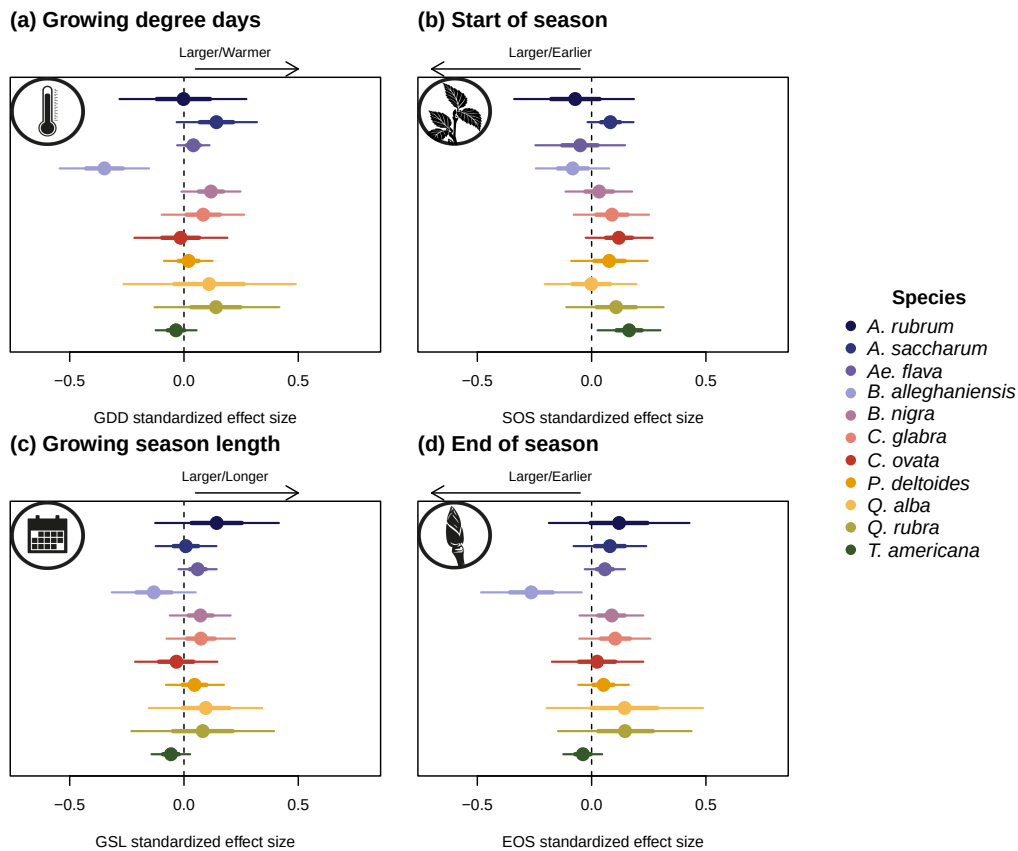


Figure S10: Effect size of each predictors across our eleven species. We depict ring width (on the log scale) change per scaled (z-scored; see data analyses section for more details) predictor for each species. We report estimates as mean and 90% uncertainty intervals (in square brackets) across our four different predictors (GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season).

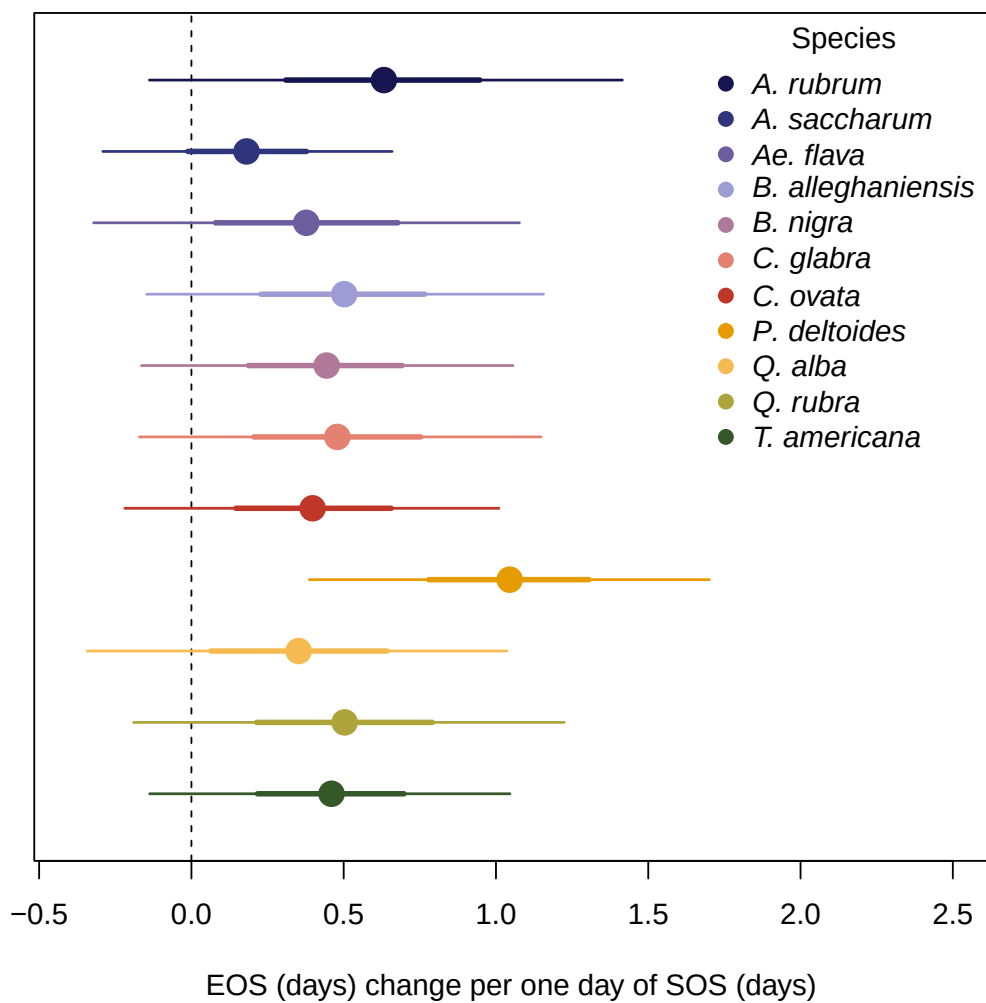


Figure S11: Effect of the start of season (SOS) on the end of season (EOS) across our eleven species. We report these slope estimates as mean and 90% uncertainty intervals.

References

Denny, E.G., Gerst, K.L., Miller-Rushing, A.J., Tierney, G.L., Crimmins, T.M., Enquist, C.A.F., Guertin, P., Rosemartin, A.H., Schwartz, M.D., Thomas, K.A. & Weltzin, J.F. (2014) Standardized phenology monitoring methods to track plant and animal activity for science and resource management applications. *International Journal of Biometeorology* **58**, 591–601.